

2.2 FORMATTING TEXTUAL DATA

The Essentials of Character Coding & Binary Transformation

| TEXTUAL DATA

Textual Data refers to a type of data that is stored and managed within computer systems, consisting of letters, numbers, and symbols that humans can read and understand.

Characteristics of Textual Data

- ❖ **Unstructured Data:** Unlike Excel tables that are neatly formatted into specific rows and columns, most textual data exists as paragraphs and notes, classifying it as unstructured data.
- ❖ **Human-Centric:** Unlike photos or audio files, it consists of natural language written specifically for human-to-human communication.

TEXTUAL DATA

Examples of Textual Data

The following items that we encounter in our daily lives are all examples of textual data:

- ❖ Emails and chat messages.
- ❖ Social media posts and comments (e.g., Facebook, X).
- ❖ News articles, books, and PDF documents.
- ❖ Product reviews (e.g., feedback written by customers about a product).

| How Do Computers Understand Textual Data?

Computers do not directly understand characters the way humans do. Therefore, textual data must go through a few stages to be processed:

- ❖ **Encoding:** Converting characters into numerical values (Binary) that a computer can understand (e.g., Unicode or ASCII systems).
- ❖ **NLP (Natural Language Processing):** An AI technology that enables computers to understand the meaning behind human text and identify the underlying sentiment.

USE CASES OF TEXTUAL DATA

Today, textual data is being used to build powerful technological solutions:

Use Case	Description
Sentiment Analysis	AI automatically analyzes customer review texts to determine whether they are satisfied or not.
Chatbots & AI	Systems like ChatGPT are trained on massive amounts of textual data to respond naturally like a human.
Search Engines	When you type a search query into Google, it searches for and displays websites containing relevant text.

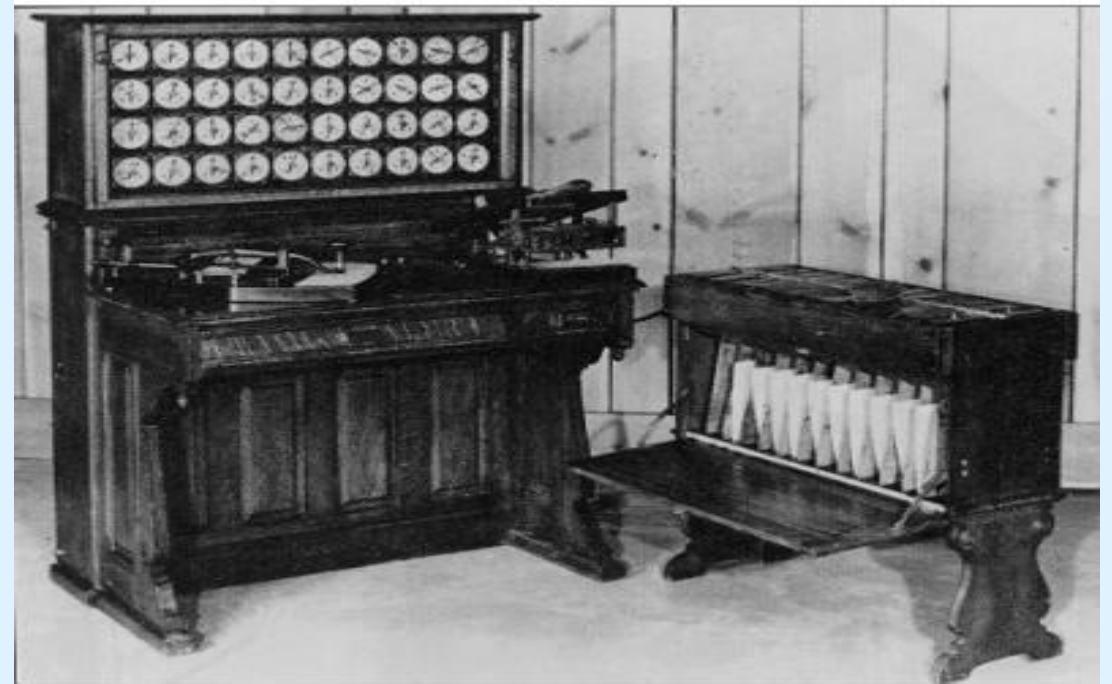
Textual Data is one of the most critical types of data in the tech world. It serves as the foundation that enables AI to draw conclusions, translate, and respond based on the text we write.

WHAT IS CHARACTER CODING?

TEXTUAL TO DIGITAL

Most communicated data is originally textual or analog. Character coding is the step that transforms alphanumeric text into binary digits (bits).

This transformation is necessary for digital transmission and processing. Without coding, computers cannot interpret human language, symbols, or numerical characters.



COMMON STANDARDS



ASCII

American Standard Code for
Information Interchange.



EBCDIC

Extended Binary Coded
Decimal Interchange Code.



LEGACY

Baudot and Hollerith formats
for early systems.

7-BIT ASCII STANDARD

128 CHARACTERS

The standard ASCII uses a 7-bit architecture, providing capacity for 128 unique characters including uppercase, lowercase, numbers, and symbols.

- ✓ Universally compatible
- ✓ Efficient for basic text

EBCDIC FORMAT

IBM CORE STANDARD

Developed for IBM mainframes, EBCDIC is an 8-bit code that can represent up to 256 unique characters.

It was designed to be an extension of the older Binary Coded Decimal (BCD) format used on punch card peripherals.

SERIAL TRANSMISSION LOGIC

1
THE LEAD BIT

SIGNIFYING ORDER

In serial transmission, the bit numbers signify the sequence. Bit number 1 is the **first signaling element** transmitted.

This chronological order ensures the receiver reconstructs the data correctly.

CODE MODIFICATIONS: ERROR DETECTION

ADDING THE 8TH BIT

Standard 7-bit ASCII can be modified to include an 8th bit for error detection purposes (Parity Checking).

This extra bit helps verify that data has not been corrupted during the communication process.

$$7 \text{ bits (Data)} + 1 \text{ bit (Parity)} = 8 \text{ bits}$$

CODE MODIFICATIONS: TRUNCATION

6-BIT TRUNCATED ASCII

Sometimes the 7-bit code is truncated to a 6-bit version to save bandwidth in specialized transmissions.

- ➖ **Reduced Capacity:** Only 64 characters instead of 128.
- ⚡ **High Efficiency:** Ideal for systems requiring limited character sets.

ENCODING COMPARISON

Format	Standard Bits	Max Characters	Common Usage
ASCII	7-Bit	128	Personal Computers, Web
EBCDIC	8-Bit	256	IBM Mainframes
Truncated ASCII	6-Bit	64	Specialized Data Links
Unicode	Variable (8-32)	1,000,000+	Universal Global Support

CODING EVOLUTION



1890S

Hollerith Cards



1960S

ASCII & EBCDIC



1980S

Extended ASCII



1990S

Unicode Launch

KEY TAKEAWAYS

TRANSFORMATION

Converting text to binary is the bedrock of digital data communication.

STANDARDS

ASCII (PC) and EBCDIC (Mainframe) are the most significant historical formats.

FLEXIBILITY

Code lengths can be modified to balance between error detection and efficiency.

2.3 MESSAGES, CHARACTERS, AND SYMBOLS

- ❖ Text messages contain alphanumeric characters.
- ❖ Characters are converted into binary bits before transmission.
- ❖ The resulting sequence is called a bit stream.
- ❖ Bits are grouped into symbols for efficient communication.

MESSAGE AND CHARACTER

- ❖ Message: Information to be transmitted.
- ❖ Example: HELLO
- ❖ Characters: H, E, L, L, O

| BIT AND BIT STREAM

- ❖ Bit: Smallest unit of digital information (0 or 1).
- ❖ Example: A = 01000001
- ❖ Bit Stream: 0100000101000010

| SYMBOLS

- ❖ A symbol represents one or more bits.
- ❖ Example: 00110110
- ❖ Grouping: 00 | 11 | 01 | 10

| M-ary Communication System

- ❖ Formula: $M = 2^k$
- ❖ M = Number of symbols
- ❖ k = Bits per symbol

Binary System ($k=1$)

- $M = 2$
- $0 \rightarrow S_0$
- $1 \rightarrow S_1$
- Uses two waveforms.

| Quaternary System (K=2)

- $M = 4$
- $00 \rightarrow S0$
- $01 \rightarrow S1$
- $10 \rightarrow S2$
- $11 \rightarrow S3$

BINARY Vs 4-ary

- Binary: 1 bit/symbol, 2 symbols
- 4-ary: 2 bits/symbol, 4 symbols
- Higher efficiency in 4-ary systems.

Digital Waveforms

- Binary:
 - 0 -> Waveform A
 - 1 -> Waveform B
- 4-ary:
 - 00 -> A, 01 -> B, 10 -> C, 11 -> D

EXAMPLE

- Message: HI
- ASCII:
- H = 01001000
- I = 01001001
- Bit Stream:
- 0100100001001001

Messages contain characters.

Characters are encoded into bits.

Bits form a bit stream.

Symbols are transmitted as digital waveforms.